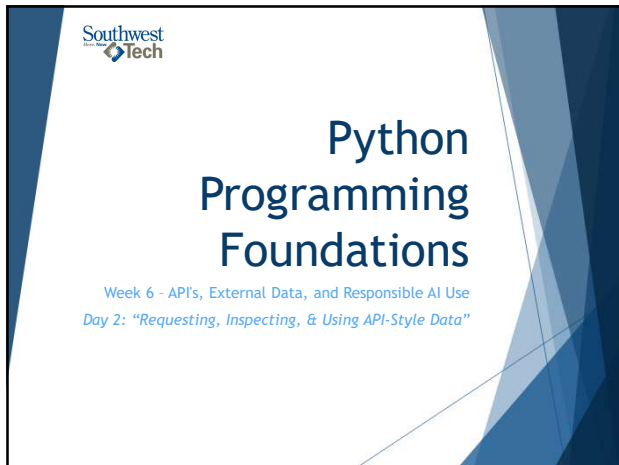
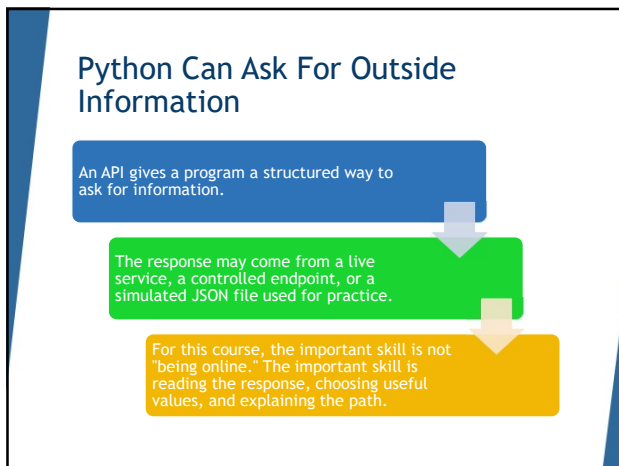
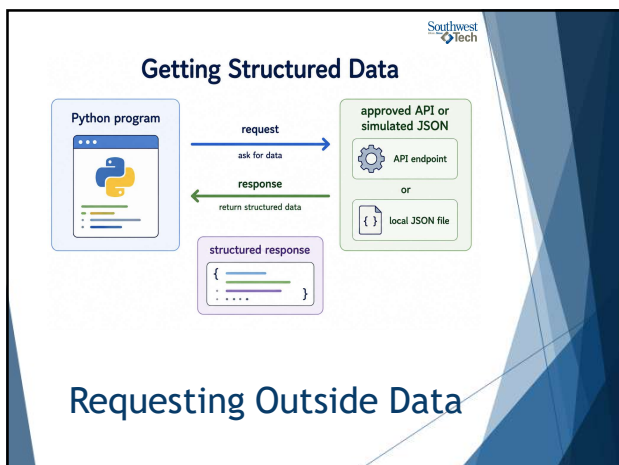




1



2



3



Review: Request, Response, Inspect, Use

The basic API-style pattern is:

- ▶ request or load data
- ▶ receive or open a response
- ▶ inspect the response shape
- ▶ select useful values
- ▶ display a clear result

4



API-Style Pattern

A simple five-step flow for working with outside data.

- 1 request or load data**
Ask for data from an endpoint or load it from a local file.
- 2 receive response**
Get the response as structured data.
- 3 inspect shape**
Look at the keys, sections, and nesting to understand it.
- 4 select values**
Choose only the values you need for your goal.
- 5 display result**
Show the result in a clear, useful way.

★ Focus on understanding the data path, not the technology behind it.

API Style Pattern

5



What Counts as Success Today

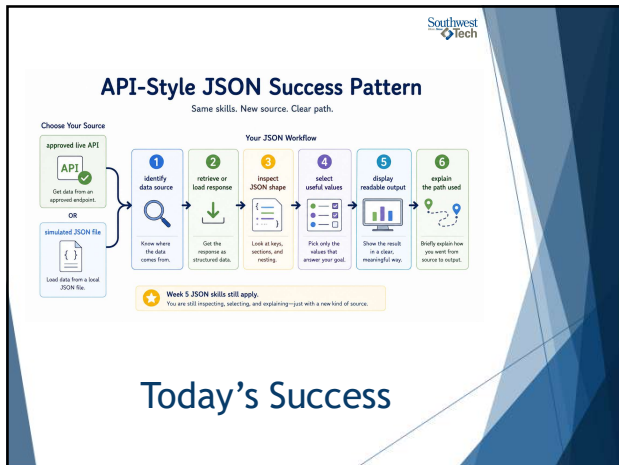
The main points:

identify the data source	retrieve or load the response	inspect the JSON shape	select useful values	display readable output	explain the path used
--------------------------	-------------------------------	------------------------	----------------------	-------------------------	-----------------------

↓

If you can explain the path from source to output, the API work becomes much less mysterious

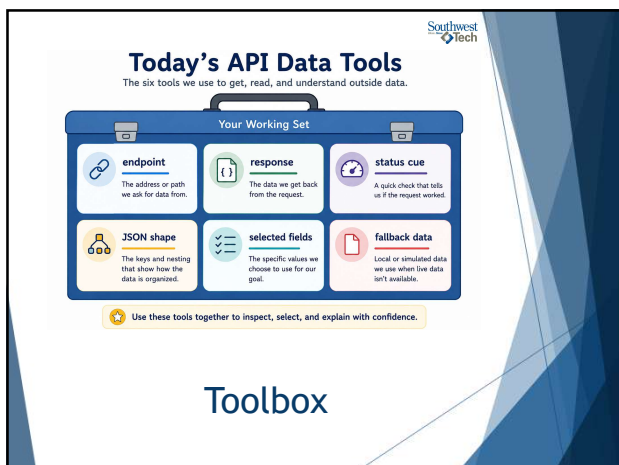
6



7



8



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What We Will “Skip” For Now

- ▶ API authentication
- ▶ creating endpoints
- ▶ API deployment
- ▶ full web-framework setup
- ▶ complex rate limits and production monitoring

*Today we consume or inspect data.
We do not build the API.*

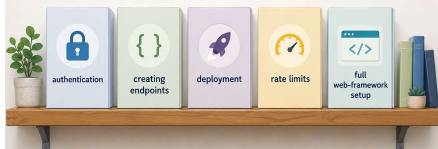


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Not Today’s Build Target

...



Today we consume or inspect data. We do not build the API.

Parked For Later

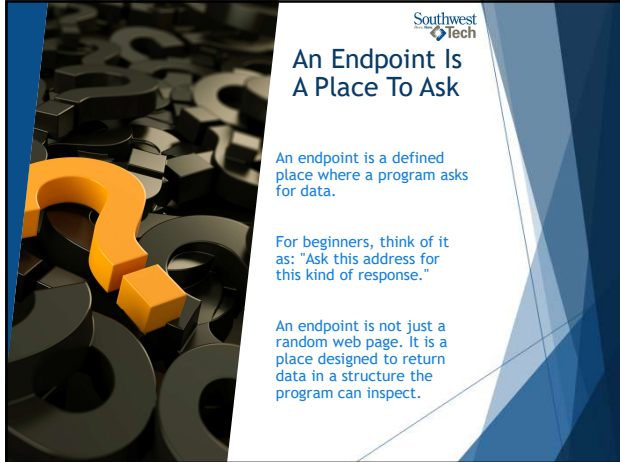
11

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API's

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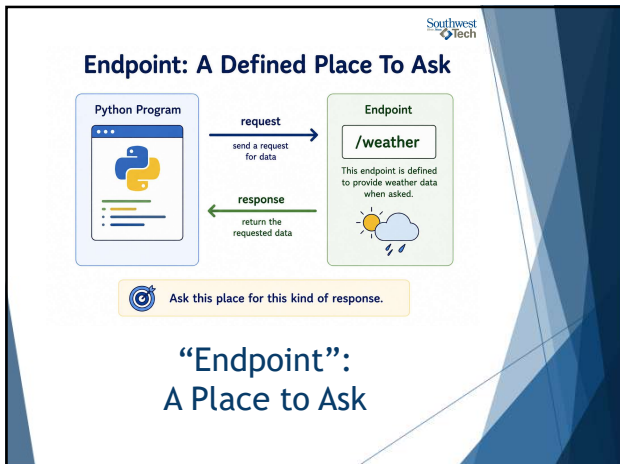
An Endpoint Is A Place To Ask

An endpoint is a defined place where a program asks for data.

For beginners, think of it as: "Ask this address for this kind of response."

An endpoint is not just a random web page. It is a place designed to return data in a structure the program can inspect.

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Endpoint: A Defined Place To Ask

Python Program

request
send a request for data

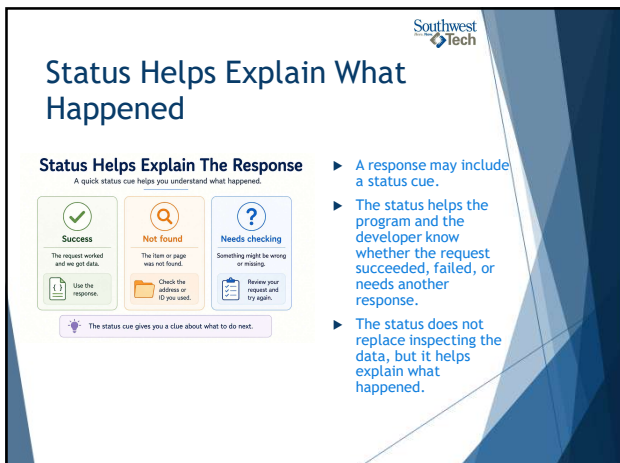
Endpoint
/weather
This endpoint is defined to provide weather data when asked.

response
return the requested data

Ask this place for this kind of response.

“Endpoint”: A Place to Ask

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Status Helps Explain What Happened

Status Helps Explain The Response

A quick status cue helps you understand what happened.

Success	Not found	Needs checking
The request worked and we got data.	The item or page was not found.	Something might be wrong or missing.
Use the response.	Check the address or ID you used.	Review your request and try again.

The status cue gives you a clue about what to do next.

- ▶ A response may include a status cue.
- ▶ The status helps the program and the developer know whether the request succeeded, failed, or needs another response.
- ▶ The status does not replace inspecting the data, but it helps explain what happened.

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Common HTTP Response Status Codes

A quick guide to what the most common status codes mean.

 2xx Success Examples: 200 OK Meaning: The request worked.	 3xx Redirect Examples: 301 / 302 Redirect Meaning: The resource moved or the request was redirected.	 4xx Client-Side Problem Examples: • 400 Bad Request • 401 Unauthorized • 403 Forbidden • 404 Not Found Meaning: The request needs to be fixed, authenticated, allowed, or aimed at the right resource.	 5xx Server-Side Problem Examples: • 500 Internal Server Error • 503 Service Unavailable Meaning: The server or service failed or is temporarily unavailable.
---	--	---	---

💡 Status codes help us decide whether to use, retry, explain, or fall back.

Status Codes

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Common HTTP Response Status Codes

- 2xx - Success**
 - 200 OK: The request worked.
- 3xx - Redirect**
 - 301/302 Redirect: The resource moved or the request was redirected.
- 4xx - Client-side problem**
 - 400 Bad Request: The request was not formed correctly.
 - 401 Unauthorized: Authentication is required.
 - 403 Forbidden: The request is understood, but access is not allowed.
 - 404 Not Found: The requested resource was not found.
- 5xx - Server-side problem**
 - 500 Internal Server Error: The server failed while processing the request.
 - 503 Service Unavailable: The service is temporarily unavailable.

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Inspect First, Extract Second



Do not guess field names.

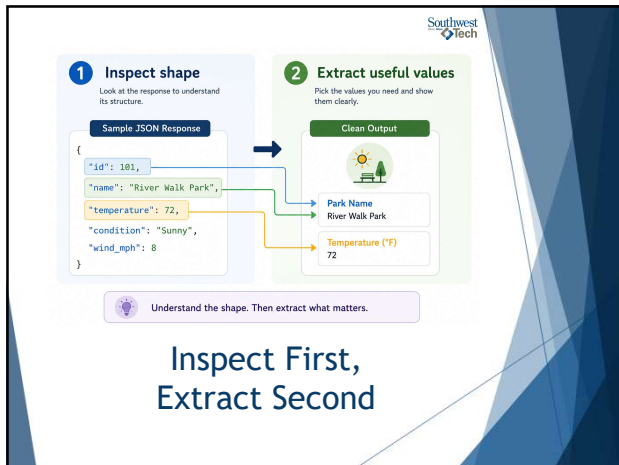


First inspect the response shape. Then select the values your program actually needs.



This is the same habit from JSON and dictionaries: look at the labels, find the path to the value, and then write code that matches the actual structure.

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Raw JSON Is Not The Finished Result

Printing raw JSON proves that data exists.

It does not prove that the program understood or used the data well.

A better result selects a few meaningful values and displays them in a way a person can read.

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Raw JSON vs Useful Output

raw response

```
{
  "city": "Madison",
  "temp": 72,
  "condition": "Sunny"
}
```

selected result

City: Madison

Temp: 72

Useful output shows that the program understood the response.

Raw JSON Is Not The Finished Result

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Live API And Simulated JSON

Two valid paths can teach the same core skill:

- ▶ **Live API path:** request data from an approved source
- ▶ **Simulated path:** load instructor-provided JSON with the same response shape

Both require inspection, selection, and explanation.

If the live path is unstable, the simulated path still lets you demonstrate the same learning target.

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Live API

real response

```
{  "city": "Madison",  "temp": 72,  "condition": "Sunny"}
```

inspect
look at the
structure

select
choose the
values you need

explain
show what
you found



Simulated JSON

stable practice response

```
{  "city": "Madison",  "temp": 72,  "condition": "Sunny"}
```

inspect
look at the
structure

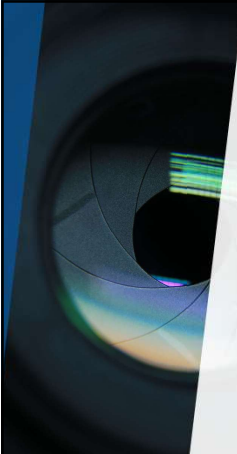
select
choose the
values you need

explain
show what
you found

 Same skill: inspect, select, explain

Live API & Simulated JSON

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Demo #1

"Parse A Simulated Response"

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What to Watch For

- ▶ response structure
- ▶ labels
- ▶ nested values if present
- ▶ selected output

Q2 Q3

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Parse A Simulated Response

simulated_weather_response.json

```
{
  "city": "Madison",
  "temp": 72,
  "condition": "Sunny",
  "humidity": 48,
  "wind_mph": 8
}
```

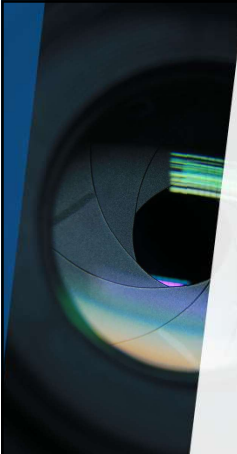
inspect structure

Labels found
We identified the keys in the response.

values ready to select
We can now choose the values we need.

Demo 1: Parse A Simulated Response

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Demo #2

"Select Values From API Data"

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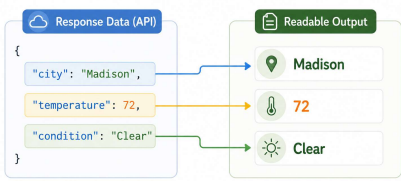
What to Watch For

- ▶ The program should choose useful values and show them clearly.
- ▶ The selected output should be easier to understand than the original response.
- ▶ Ask yourself: *what did my program make clearer than the raw data?*

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Select Values From API Data



```
{
  "city": "Madison",
  "temperature": 72,
  "condition": "Clear"
}
```

Madison
72
Clear

Demo 2: Select Values From API Data

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Demo #3

"Error And Fallback Path"

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What to Watch For

- ▶ Responsible API-style programs plan for problems.
- ▶ The fallback path keeps the learning target alive when the live path is not available.
- ▶ Fallback is not a weaker path. It is a way to preserve the same skill when the outside service is unavailable or unpredictable.

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Error And Fallback Path

1 Live Path Unavailable



The live API cannot be reached or is not responding.

2 Use Fallback JSON

Fallback JSON

Use a local, stable JSON file to continue practicing and testing your code.

```
JSON
{
  "city": "Madison",
  "temp": 72, "condition": "Clear"
}
```

Same Goal, Same Skill



Inspect

Look at the data structure.



Select

Choose the values you need.



Explain

Show what you found and why it matters.

Demo 3: Error And Fallback Path

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Assignment #11

For Assignment 11, build a small API-style data fetcher.

Your result should

- ▶ retrieve or load response data
- ▶ select useful values
- ▶ display them clearly
- ▶ and explain your data path

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Assignment 11 API Data Fetcher

1 choose data source

Pick a live API or a simulated JSON file.

Live API
Simulated JSON file

2 inspect response

Look at the response structure and identify the labels.

```
{
  "city": "Madison",
  "temp": 72,
  "condition": "Clear"
}
```

3 select useful values

Choose the values that answer your goal or question.

city: Madison
temp: 72
condition: Clear

4 explain data path

Explain where the data came from, what you selected, and why.

- Source: Open-Meteo API
- Path: city, temp, condition
- Why: shows current weather for a city

Goal: turn an API response into useful, explained information.

Assignment 11

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Evidence For A11 & A12

Your submission should show:

- ▶ code file
- ▶ data source or simulated JSON path
- ▶ selected-value output
- ▶ short request/response explanation
- ▶ validation or error note
- ▶ AI-use note if AI materially helped

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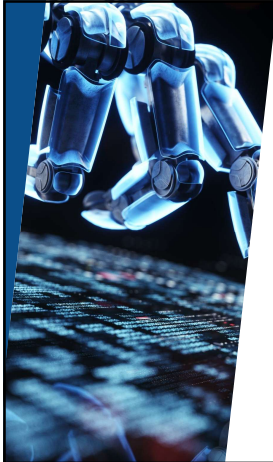
Assignment 11 Evidence

- ☐ **1 code file**
Your Python file that runs and fetches or reads the data.
- ☐ **2 data source or simulated path**
The API endpoint used or the local simulated JSON file path.
- ☐ **3 selected output**
The values you extracted and displayed clearly.
- ☐ **4 request/response explanation**
What was requested, what was returned, and how the data was used.
- ☐ **5 validation note**
How you checked that the data and output are correct.
- ☐ **6 AI-use note, if used**
What AI tool was used and how it supported your work.

★ Clear evidence shows your process and understanding.

Evidence

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AI Use In Lab 0n

- ▶ AI can suggest API code that looks believable but does not match the actual response.
- ▶ You must inspect the response, run the code, and explain what human decision you made.
- ▶ If AI names a field that does not exist in your response, your program is still wrong even if the code looks professional.

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AI suggestion

```
</>
```

Code or approach suggested by AI

→

human verification

- ✓ Understand the goal
- ✓ Confirm the data
- ✓ Choose what to use

→

Actual response shape

```
{
  "city": "Madison",
  "temp": 70,
  "condition": "Clear"
}
```

The real JSON response returned by the API

Run it, inspect it, explain it.

Your understanding is what turns data into useful information.

AI-Assisted Code Must Be Verified

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Closing Success Check

By the end of today, you should be able to say:

"I know where the data came from, what shape it had, which values I selected, and how I verified the result."

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